

CHUQUR O'RGANISH USULLARI · Ma'ruza konspekti (Lecture Note)

3-ma'ruza. Konvolyutsion neyron tarmoqlar (CNN)

2026–2027 o'quv yili

Reja. 1) To'liq bog'langan qatlamning tasvirdagi kamchiliklari. 2) Konvolyutsiya (kross-korrelyatsiya) va ekvivariantlik. 3) Chiqish o'lchami, padding, stride, dilation. 4) Parametrlar va hisoblash hajmi; 1×1 va depthwise separable konvolyutsiya. 5) Pooling va qabul maydoni. 6) Konvolyutsiya orqali backpropagation. 7) Batch normalization va dropout. 8) LeNet, AlexNet, VGG, ResNet. 9) Python (NumPy, PyTorch). 10) Nazorat savollari va mashqlar.

O'quv natijalari. Ma'ruzadan so'ng talaba: ko'p kanalli konvolyutsiyani qo'lda va dasturda hisoblaydi; chiqish o'lchami, parametrlar soni, FLOP va qabul maydonini topadi; konvolyutsiyaning siljishga ekvivariantligini isbotlaydi; konvolyutsiya va pooling orqali gradientlarni chiqaradi; batch normalization formulalarini qo'llaydi; klassik arxitekturalarning g'oyalarini tushuntiradi va PyTorch'da CNN quradi.

Tayanch iboralar. konvolyutsiya, kross-korrelyatsiya, filtr (yadro), xususiyat xaritasi (feature map), kanal, padding, stride, dilation, lokal bog'lanish, og'irliklarni bo'lishish (weight sharing), ekvivariantlik, max-pooling, average pooling, global average pooling, qabul maydoni (receptive field), im2col, batch normalization, residual bog'lanish.

1. Muammo: to'liq bog'langan qatlam tasvirda

$224 \times 224 \times 3$ rangli tasvirni 1000 ta neyronga to'liq ulash uchun $224 \cdot 224 \cdot 3 \cdot 1000 = 150\,528\,000$ og'irlik (va 1000 siljish) kerak — bitta qatlam uchun. Bundan ham muhimi, bunday qatlam har bir piksel o'rniga alohida og'irlik beradi: chap yuqori burchakdagi qirrani taniydigan og'irliklar o'ng pastki burchakdagi xuddi shunday qirra haqida hech narsa “bilmaydi”. 3×3 o'lchamli 64 ta filtrlil konvolyutsion qatlam esa $3 \cdot 3 \cdot 3 \cdot 64 + 64 = 1\,792$ parametrga ega va har bir filtrni butun tasvir bo'ylab qo'llaydi ($150\,528\,000/1\,792 = 84\,000$ marta kam).

Tasvirlar uchun uchta farazdan (induktiv siljish) foydalaniladi:

1. *Lokallik*: piksel ma'nosi asosan uning yaqin qo'shnilariga bog'liq — filtr kichik (3×3).
2. *Og'irliklarni bo'lishish*: bir xil belgi (qirra, burchak) istalgan joyda paydo bo'lishi mumkin — bitta filtr hamma joyda qo'llanadi.
3. *Ierarxiya*: murakkab belgilar oddiylaridan tuziladi — qatlamlar ketma-ketligi (qirra $\rightarrow$ tekstura $\rightarrow$ qism $\rightarrow$ obyekt).

Bu farazlar jadval ma'lumotlari uchun o'rinsiz (ustunlar tartibi ma'nosiz), shuning uchun CNN asosan tasvir, audio-spektrogramma, vaqt qatorlari kabi “panjara” ma'lumotlarda ishlatiladi.

2. Konvolyutsiya

Ta'rif 2.1 (Konvolyutsion qatlam). Kirish $X \in \mathbb{R}^{C_{in} \times H \times W}$, filtrlar $\mathcal{W} \in \mathbb{R}^{C_{out} \times C_{in} \times K_h \times K_w}$, siljishlar $b \in \mathbb{R}^{C_{out}}$, qadam (stride) s , to'ldirish (padding) p , kengaytirish (dilation) d bo'lsin. Chiqish:

$$Y_{o,i,j} = b_o + \sum_{c=1}^{C_{in}} \sum_{u=0}^{K_h-1} \sum_{v=0}^{K_w-1} \mathcal{W}_{o,c,u,v} \tilde{X}_{c, is+ud, js+vd},$$

bunda $\tilde{X}$ — har tomondan p qator/ustun nol bilan to'ldirilgan X (0 dan boshlanuvchi indekslar). $Y_{o,:}$ — *o*-xususiyat xaritasi.

Har bir chiqish kanali barcha kirish kanallarini jamlaydi: 3×3 filtr rangli tasvirda aslida $3 \times 3 \times 3$ “kub”. Matematik konvolyutsiyada yadro ag'dariladi, $(X * K)_{i,j} = \sum_{u,v} K_{u,v} X_{i-u, j-v}$; kutubxonalaridagi `Conv2d` esa ag'darmaydi, ya'ni *kross-korrelyatsiya* hisoblaydi. Filtr o'rganilsa, farq yo'q (tarmoq ag'darilgan filtrni o'rganadi), lekin qo'lda berilgan filtrlarda (Sobel) natija ishorasi o'zgarishi mumkin.

Misol 2.2 (Qo'lda hisoblash). $X = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 2 & 1 & 0 \end{pmatrix}$, $K = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, $s = 1$, $p = 0$, $b = 0$.

Yechim. Chiqish 2×2 . Har bir oynada “chap yuqori minus o'ng past”: $Y_{00} = 1 - 1 = 0$, $Y_{01} = 2 - 3 = -1$, $Y_{10} = 0 - 1 = -1$, $Y_{11} = 1 - 0 = 1$, ya'ni $Y = \begin{pmatrix} 0 & -1 \\ -1 & 1 \end{pmatrix}$. Haqiqiy konvolyutsiyada ag'darilgan yadro $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} = -K$, shuning uchun bu misolda natija $-Y$ bo'ladi. $p = 1$ bilan chiqish 4×4 : $\begin{pmatrix} -1 & -2 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ -2 & -1 & 1 & 3 \\ 0 & 2 & 1 & 0 \end{pmatrix}$ (chetdagi elementlarda oynaning bir qismi nol to'ldirishga tushadi). $\triangleleft$

$$1 \cdot 1 + 2 \cdot 0 + 0 \cdot 0 + 1 \cdot (-1) = 0$$

$X (3 \times 3)$ $K (2 \times 2)$ $Y (2 \times 2)$

Misol 2.3 (Ikki kanalli kirish). $X_1 = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 1 & 2 & 0 \end{pmatrix}$, $X_2 = \begin{pmatrix} 0 & 1 & 1 \\ 2 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$, bitta filtr $\mathcal{W}_1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $\mathcal{W}_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $b = 1$.

Yechim. $Y_{00} = \underbrace{(1+1)}_{c=1} + \underbrace{(1+2)}_{c=2} + 1 = 6$, $Y_{01} = (0+1) + (1+0) + 1 = 3$, $Y_{10} = (0+2) + (0+1) + 1 = 4$, $Y_{11} = (1+0) + (1+1) + 1 = 4$; $Y = \begin{pmatrix} 6 & 3 \\ 4 & 4 \end{pmatrix}$. Bitta filtr ikki kanaldan *bitta* xarita hosil qiladi; C_{out} ta filtr — C_{out} ta xarita. Parametrlar: $2 \cdot 2 \cdot 2 + 1 = 9$. $\triangleleft$

Teorema 2.4 (Siljishga ekvivariantlik). $(T_{\mathbf{a}}X)_{i,j} = X_{i-a_1,j-a_2}$ — siljish operatori bo'lsin ($\mathbb{Z}^2$ da aniqlangan, cheklangan tayanchli X). $s = 1$ konvolyutsiya $\mathcal{C}$ uchun $\mathcal{C}(T_{\mathbf{a}}X) = T_{\mathbf{a}}\mathcal{C}(X)$.

Isbot. $\mathcal{C}(T_{\mathbf{a}}X)_{i,j} = b + \sum_{u,v} K_{u,v} X_{i+u-a_1,j+v-a_2} = \mathcal{C}(X)_{i-a_1,j-a_2} = (T_{\mathbf{a}}\mathcal{C}(X))_{i,j}$. $\square$

Ya'ni belgi siljisa, xususiyat xaritasidagi javob ham shuncha siljiydi. Chekli tasvirda teorema chegaralardan uzoqda o'rinli; $s > 1$ da faqat s ga karrali siljishlar uchun. *Invariantlik* (javob umuman o'zgarmasligi) keyin global pooling orqali olinadi.

Amaliy talqin: Sobel filtri — qo'lda berilgan qirra detektori

$G_x = \begin{pmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{pmatrix}$, $G_y = G_x^T$. Chap yarmi qora (0), o'ng yarmi oq (1) bo'lgan 4×4 tasvirga (ustunlar 0, 0, 1, 1) `Conv2d` bilan qo'llansak, 2×2 chiqishning barcha elementlari +4: vertikal qirra topildi. Bir jinsli sohada chap va o'ng ustunlar bir-birini yo'qotadi va javob 0. Ag'darilgan (haqiqiy) konvolyutsiya -4 beradi. CNN ning birinchi qatlami aynan shunday filtrlarni ma'lumotdan o'zi o'rganadi; sanoatda bu — sifat nazorati (yoriq, dog'), tibbiy tasvirlarda chegara aniqlash.

3. Chiqish o'lchami

Teorema 3.1. Kirish balandligi H , yadro K , padding p , stride s , dilation d bo'lsa,

$$H_{\text{out}} = \left\lfloor \frac{H + 2p - d(K - 1) - 1}{s} \right\rfloor + 1.$$

Isbot. Kengaytirilgan yadroning samarali kengligi $K_e = d(K - 1) + 1$ (elementlar orasida $d - 1$ ta bo'shliq). To'ldirilgan kenglik $H + 2p$. i -chiqish oynasi is dan boshlanadi va u $is + K_e - 1 \leq H + 2p - 1$ bo'lganda to'liq ichkarida, ya'ni $i \leq (H + 2p - K_e)/s$. $i = 0, 1, \dots$ bo'lgani uchun oynalar soni $\lfloor (H + 2p - K_e)/s \rfloor + 1$. $\square$

H	K	p	s	d	H_{out}	PyTorch	Izoh
32	3	1	1	1	32	32	“same”: $p = (K - 1)/2$, $s = 1$
32	3	1	2	1	16	16	stride 2 — o'lcham yarmiga
32	3	2	1	2	32	32	dilation: $K_e = 5$
28	5	0	1	1	24	24	“valid”
28	5	2	2	1	14	14	$\lfloor 13,5 \rfloor + 1$
224	7	3	2	1	112	112	ResNet birinchi qatlami
28	3	0	1	3	22	22	$K_e = 7$

Barcha qatorlar `torch.nn.functional.conv2d` bilan tekshirilgan. Xuddi shu formula pooling qatlamlariga ham qo'llanadi (masalan, $112 \rightarrow 56$: $K = 3$, $p = 1$, $s = 2$).

4. Parametrlar, hisoblash hajmi va maxsus konvolyutsiyalar

Xossa 4.1. Guruhlanmagan konvolyutsion qatlamda parametrlar va ko'paytirish-qo'shish amallari (MAC) soni

$$C_{\text{out}}(C_{\text{in}}K_hK_w + 1), \quad \text{MAC} \approx N H_{\text{out}}W_{\text{out}} C_{\text{out}}C_{\text{in}}K_hK_w$$

(N — paket hajmi). Parametrlar tasvir o'lchamiga bog'liq emas, hisob esa bog'liq.

Misol 4.2. $C_{\text{in}} = 64$, $C_{\text{out}} = 128$, 3×3 , chiqish 56×56 , $N = 1$.

Yechim. Parametrlar $128(64 \cdot 9 + 1) = 73\,856$; $\text{MAC} = 56 \cdot 56 \cdot 128 \cdot 64 \cdot 9 = 231\,211\,008 \approx 2,3 \cdot 10^8$. Faollashtirishlar: $56 \cdot 56 \cdot 128 = 401\,408$ son — parametrlardan 5 marta ko'p. Orqaga yurish uchun ular saqlanadi (2-ma'ruza), shuning uchun CNN o'qitish xotirasini asosan faollashtirishlar belgilaydi. $\triangleleft$

1×1 **konvolyutsiya** har bir pikselda kanallarni chiziqli aralashtiradi (piksel bo'yicha to'liq bog'langan qatlam): $64 \rightarrow 128$ uchun $128 \cdot 65 = 8\,320$ parametr (3×3 dan 8,9 marta kam). U kanallar sonini kamaytirish/oshirish uchun ishlatiladi (ResNet “bottleneck”, Inception).

Depthwise separable konvolyutsiya (MobileNet): avval har bir kanal alohida 3×3 filtr bilan (`groups=C_in`), keyin 1×1 bilan aralashtiriladi. $64 \rightarrow 128$: $(64 \cdot 9 + 64) + (64 \cdot 128 + 128) = 640 + 8\,320 = 8\,960$ parametr — oddiy 3×3 dan 8,2 marta kam; MAC nisbati $\frac{1}{C_{\text{out}}} + \frac{1}{K^2} \approx 0,119$.

5. Pooling va qabul maydoni

Max-pooling oynadagi eng katta qiymatni, **average pooling** o'rtachani oladi (odatda 2×2 , $s = 2$: o'lcham yarmiga, kanallar soni o'zgarmaydi, parametr yo'q). **Global average pooling** (GAP) har bir xaritani bitta songa aylantiradi: $C \times H \times W \rightarrow C$ — klassifikatordan oldin `Flatten+Linear` ning og'ir parametrlarini almashtiradi va istalgan kirish o'lchamida ishlaydi.

Gradient. Max-pooling: $\partial L / \partial x$ faqat oynadagi maksimal elementga $\bar{y}$, qolganlariga 0 (ReLU ga o'xshash “marshrutlash”). Average ($k \times k$): har bir elementga $\bar{y}/k^2$.

Misol 5.1 (Max-pooling orqali gradient). A3-mashqdagi 4×4 matritsaga 2×2 , $s = 2$ max-pooling, yuqoridan kelgan gradient $\bar{Y} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$.

Yechim. Maksimumlar: 4 (1-qator, 0-ustun), 2 (0-qator, 2-ustun), 5 (2-qator, 2-ustun); pastki chap oynada 2 ikki marta uchraydi — PyTorch birinchisini (3-qator, 0-ustun) tanlaydi. Demak $\bar{X} = \begin{pmatrix} 0 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 3 & 0 & 0 & 0 \end{pmatrix}$ — 12 ta element gradient olmaydi. $\triangleleft$

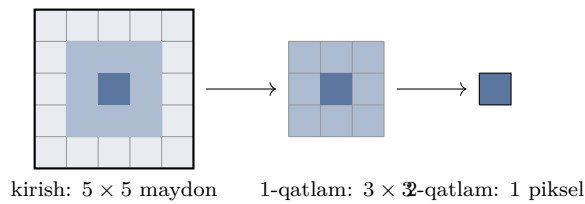
Izoh 5.2 (Pooling to'liq siljish invariantligi bermaydi). 1D misol, oyna 2, stride 2: $[0, 1, 0, 0] \rightarrow \max[1, 0]$; bir pozitsiya siljigan $[0, 0, 1, 0] \rightarrow [0, 1]$. Natija o'zgardi: pooling faqat oyna *ichidagi* kichik siljishlarga chidamli. Zamonaviy CNN larning kichik siljishlarga ham sezgir bo'lishi mumkinligi (Azulay–Weiss, 2019) shu bilan bog'liq.

Ta'rif 5.3 (Qabul maydoni). Chiqish elementiga ta'sir qiluvchi kirish sohasining o'lchami r_l . $r_0 = 1$, $j_0 = 1$ dan:

$$r_l = r_{l-1} + (k_l - 1) d_l j_{l-1}, \quad j_l = j_{l-1} s_l,$$

j_l — l -qatlamdagi qo'shni chiqishlar orasidagi kirish pikselidagi masofa.

Misollar: ikkita $3 \times 3 \Rightarrow 5$; bitta $5 \times 5 \Rightarrow 5$; uchta $3 \times 3 \Rightarrow 7$; [conv3, conv3, maxpool2 ($s = 2$), conv3, conv3] $\Rightarrow 5 \rightarrow 6 \rightarrow 10 \rightarrow 14$; $d = 1, 2, 4$ li uchta $3 \times 3 \Rightarrow 15$ (dilation qabul maydonini eksponensial o'stiradi).



Ikki ketma-ket 3×3 konvolyutsiya: chiqishdagi bitta piksel 1-qatlamdagi 3×3 va kirishdagi 5×5 sohaga bog'liq.

Xossa 5.4 (Nega kichik filtrlar ketma-ketligi (VGG)?). C kanalli ikkita 3×3 qatlam: 5×5 qabul maydoni, $18C^2$ og'irlik va 2 ta chiziqsizlik; bitta 5×5 : xuddi shu maydon, $25C^2$ og'irlik, 1 chiziqsizlik. Uchta 3×3 ($27C^2$) bitta 7×7 ($49C^2$) o'rnini bosadi.

6. Konvolyutsiya orqali backpropagation

Soddalik uchun $s = 1$, $p = 0$, $d = 1$, bitta namuna; $\delta_{o,i,j} = \partial L / \partial Y_{o,i,j}$.

Teorema 6.1.

$$\frac{\partial L}{\partial \mathcal{W}_{o,c,u,v}} = \sum_{i,j} \delta_{o,i,j} X_{c,i+u,j+v}, \quad \frac{\partial L}{\partial b_o} = \sum_{i,j} \delta_{o,i,j}, \quad \frac{\partial L}{\partial X_{c,a,b}} = \sum_o \sum_{u,v} \delta_{o,a-u,b-v} \mathcal{W}_{o,c,u,v},$$

oxirgisida δ indeksi chegaradan chiqsa, 0 olinadi.

Isbot. $Y_{o,i,j}$ ning $\mathcal{W}_{o,c,u,v}$ bo'yicha hosilasi $X_{c,i+u,j+v}$; zanjir qoidasi bo'yicha barcha (i, j) lar qo'shiladi (og'irlik bo'lishilgani uchun gradientlar yig'iladi). $X_{c,a,b}$ esa $i + u = a$, $j + v = b$ bo'lgan barcha $Y_{o,i,j}$ larga $\mathcal{W}_{o,c,u,v}$ koeffitsiyent bilan kiradi. $\square$

Ya'ni $\nabla \mathcal{W}$ — kirish va δ ning kross-korrelyatsiyasi, ∇X — nol bilan to'ldirilgan δ ning *ag'darilgan* yadro bilan korrelyatsiyasi ("to'liq" konvolyutsiya; ConvTranspose2d aynan shu amal). Stride $s > 1$ da δ elementlari orasiga $s - 1$ ta nol qo'yiladi.

Misol 6.2. 2.2-misoldagi X , K va $L = \sum_{i,j} Y_{i,j}$ (ya'ni $\delta \equiv 1$).

Yechim. $\partial L / \partial K_{u,v}$ — X ning 2×2 oynalar bo'yicha yig'indisi:

$$\frac{\partial L}{\partial K} = \begin{pmatrix} 1 + 2 + 0 + 1 & 2 + 0 + 1 + 3 \\ 0 + 1 + 2 + 1 & 1 + 3 + 1 + 0 \end{pmatrix} = \begin{pmatrix} 4 & 6 \\ 4 & 5 \end{pmatrix}, \quad \frac{\partial L}{\partial X} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & -1 \\ 0 & -1 & -1 \end{pmatrix},$$

bunda $\partial L / \partial X_{a,b}$ — (a, b) ni qamrab olgan oynalardagi mos $K_{u,v}$ lar yig'indisi (markaziy element 4 ta oynada: $1 + 0 + 0 - 1 = 0$). Ikkalasi ham PyTorch autograd bilan mos. $\triangleleft$

Misol 6.3 (Kichik CNN da to'liq oldinga va orqaga yurish). $X = \begin{pmatrix} 3 & 0 & 2 & 1 \\ 0 & 1 & 1 & 0 \\ 2 & 1 & 0 & 1 \\ 1 & 0 & 1 & 2 \end{pmatrix}$, $K = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$,

$b = 0 \rightarrow \text{ReLU} \rightarrow \text{global max-pooling } m \rightarrow z = wm + c$ ($w = 0,5$, $c = -1$) $\rightarrow \text{sigmoid} + \text{binary cross-entropy}$, $y = 1$.

Yechim. *Oldinga.* $Y = \begin{pmatrix} 4 & -2 & 0 \\ -2 & -1 & 2 \\ 0 & 2 & 0 \end{pmatrix}$ (masalan, $Y_{00} = 3 - 0 - 0 + 1$), ReLU manfiylarni nolga

aylantiradi, $m = 4$ (yagona maksimum Y_{00}), $z = 1$, $p = \sigma(1) \approx 0,7311$, $L \approx 0,3133$. *Orqaga.* $\partial L / \partial z = p - 1 \approx -0,2689$; $\partial L / \partial w = -0,2689 \cdot 4 \approx -1,0758$, $\partial L / \partial c \approx -0,2689$; $\partial L / \partial m = w(p - 1) \approx -0,1345$ faqat Y_{00} ga boradi (max-pooling marshruti, $\text{ReLU}' = 1$). Teorema 6.1 bo'yicha $\partial L / \partial K = -0,1345 \cdot \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix} \approx \begin{pmatrix} -0,4034 & 0 \\ 0 & -0,1345 \end{pmatrix}$. Gradient tushish K_{00} va K_{11} ni oshiradi — filtr aynan shu naqshga ko'proq “javob” beradigan bo'ladi. (PyTorch autograd bilan tekshirilgan.) $\triangleleft$

im2col. Har bir oyna ($C_{\text{in}}K^2$ son) matritsaning bitta ustuniga yoziladi: $X_{\text{col}} \in \mathbb{R}^{C_{\text{in}}K^2 \times H_{\text{out}}W_{\text{out}}}$; filtrlar $\mathcal{W}_{\text{row}} \in \mathbb{R}^{C_{\text{out}} \times C_{\text{in}}K^2}$. U holda $Y = \mathcal{W}_{\text{row}}X_{\text{col}} + \mathbf{b}\mathbf{1}^T$ — bitta matritsa ko'paytmasi, va 2-ma'ruzadagi $\nabla W = \dots$ formulalari to'g'ridan-to'g'ri qo'llanadi. Kutubxonalar (cuDNN) im2col, FFT yoki Winograd algoritmlaridan foydalanadi.

Transposed konvolyutsiya (ConvTranspose2d) — konvolyutsiyaning X bo'yicha gradient amali; fazoviy o'lchamni *oshiradi* (segmentatsiya, generativ modellar): $H_{\text{out}} = (H - 1)s - 2p + d(K - 1) + p_{\text{out}} + 1$. Masalan, $K = 2$, $s = 2$, $p = 0$: $H \rightarrow 2H$.

7. Batch normalization va dropout

Ta'rif 7.1 (Batch normalization (Ioffe–Szegedy, 2015)). Har bir kanal c uchun paket va fazoviy o'lchamlar ($N \times H \times W$ ta son) bo'yicha

$$\mu_c = \text{o'rtacha}, \quad \sigma_c^2 = \text{dispersiya}, \quad \hat{x} = \frac{x - \mu_c}{\sqrt{\sigma_c^2 + \epsilon}}, \quad y = \gamma_c \hat{x} + \beta_c.$$

γ_c, β_c o'rganiladi ($2C$ parametr). O'qitishda μ, σ^2 joriy paketdan olinadi va sirg'aluvchi o'rtacha yangilanadi; inferensiyada (`model.eval()`) shu saqlangan o'rtachalar ishlatiladi.

Misol: bitta kanal qiymatlari (1, 2, 3, 6) bo'lsa, $\mu = 3$, $\sigma^2 = 3,5$ ($1/N$ bilan) va

$$\hat{x} \approx (-1,069; -0,535; 0; 1,604).$$

BN o'qitishni barqarorlashtiradi, kattaroq η ga imkon beradi va yengil regularizatsiya beradi; odatiy tartib Conv $\rightarrow$ BN $\rightarrow$ ReLU (Conv ning siljishi keraksiz, `bias=False`: BN dagi β uni o'z ichiga oladi). Kichik paketlarda statistika shovqinli — GroupNorm yoki LayerNorm afzal.

Xossa 7.2 (BN orqali gradient). Bitta kanal, N ta qiymat, $\bar{y}_k = \partial L / \partial y_k$, $s = \sqrt{\sigma^2 + \epsilon}$ bo'lsa: $\frac{\partial L}{\partial \gamma} =$

$$\sum_k \bar{y}_k \hat{x}_k, \quad \frac{\partial L}{\partial \beta} = \sum_k \bar{y}_k,$$

$$\frac{\partial L}{\partial x_k} = \frac{\gamma}{Ns} \left(N\bar{y}_k - \sum_j \bar{y}_j - \hat{x}_k \sum_j \bar{y}_j \hat{x}_j \right).$$

Isbot g'oyasi. $\hat{x} = (x - \mu\mathbf{1})/s$, $\partial \mu / \partial x_k = \frac{1}{N}$, $\partial s / \partial x_k = \hat{x}_k / N$; $\partial \hat{x}_j / \partial x_k = \frac{1}{s} \left(\delta_{jk} - \frac{1}{N} - \frac{\hat{x}_j \hat{x}_k}{N} \right)$ ni $\gamma \bar{y}_j$ ga ko'paytirib j bo'yicha yig'amiz. $\square$

$x = (1, 2, 3, 6)$, $\gamma = 2$, $\bar{y} = (0,5; -1; 2; 1)$ uchun formula $\approx (0,3245; -1,5081; 1,4699; -0,2863)$ beradi — autograd bilan bir xil. Gradient yig'indisi 0: BN kanal bo'yicha umumiy siljishga befarq.

Dropout (2-ma'ruza) CNN da odatda faqat to'liq bog'langan boshda ishlatiladi (AlexNet, VGG).

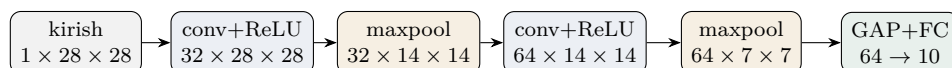
8. Klassik arxitekturalar

Arxitektura	Parametrlar	Asosiy g'oya
LeNet-5 (1998)	61 706	$2 \times (\text{conv } 5 \times 5 + \text{subsampling}) + 3 \text{ FC}$; tanh; qo'lda yozilgan raqamlar
AlexNet (2012)	61,1 mln	5 conv + 3 FC, ReLU, dropout, augmentatsiya, GPU; ILSVRC-2012 top-5 xato 15,3%
VGG-16 (2014)	138,4 mln	faqat 3×3 conv bloklari + maxpool; parametrlarning 89% i (123,6 mln) FC qatlamlarda
ResNet-18/50/152 (2015)	11,7 / 25,6 / 60,2 mln	residual bog'lanish $\mathbf{y} = \mathbf{x} + F(\mathbf{x})$, BN, GAP; ILSVRC-2015 g'olibi (top-5 3,57%, ansambl)

(Parametrlar torchvision.models bo'yicha, 1000 sinf; LeNet-5 — 28×28 kirishli zamonaviy varianti.)

Kichik CNN ning “o'lcham yo'li” (28×28 kulrang tasvir, 10 sinf):

Qatlam	Chiqish o'lchami	Parametrlar
Conv2d(1,32,3,padding=1) + ReLU	$32 \times 28 \times 28$	$32(9 + 1) = 320$
MaxPool2d(2)	$32 \times 14 \times 14$	0
Conv2d(32,64,3,padding=1) + ReLU	$64 \times 14 \times 14$	$64(288 + 1) = 18\,496$
MaxPool2d(2)	$64 \times 7 \times 7$	0
AdaptiveAvgPool2d(1) + Flatten	64	0
Linear(64,10)	10	650
Jami		19 466



Tipik naqsh: fazoviy o'lcham kamayadi ($28 \rightarrow 14 \rightarrow 7 \rightarrow 1$), kanallar soni oshadi ($1 \rightarrow 32 \rightarrow 64$). GAP o'rniga Flatten+Linear(3136,10) qo'yilsa, bosh 31 370 parametrga ega bo'lardi.

Teorema 8.1 (Residual bog'lanish va gradient). $\mathbf{y} = \mathbf{x} + F(\mathbf{x})$ bo'lsa, $\frac{\partial \mathbf{y}}{\partial \mathbf{x}} = I + \frac{\partial F}{\partial \mathbf{x}}$ va $\bar{\mathbf{x}} = \bar{\mathbf{y}} + \left(\frac{\partial F}{\partial \mathbf{x}}\right)^T \bar{\mathbf{y}}$. L ta blok uchun $\bar{\mathbf{x}}_0 = \prod_l (I + J_l)^T \bar{\mathbf{x}}_L = \bar{\mathbf{x}}_L + \sum_l J_l^T \bar{\mathbf{x}}_L + \dots$ — gradientning to'g'ridan-to'g'ri (identik) yo'li bor.

Oddiy tarmoqda $\bar{\mathbf{x}}_0 = \prod_l J_l^T \bar{\mathbf{x}}_L$: skalyar holatda $J_l = 0,1$ bo'lsa, 10 qatlamdan keyin 10^{-10} ; residual tarmoqda $1,1^{10} \approx 2,59$. Shu sababli 152 qatlamli ResNet o'qitiladi, oddiy chuqur tarmoq esa chuqurlik oshganda *o'quv* xatosi ham yomonlashadi (degradatsiya muammosi, He va boshq., 2016). Agar optimal $F \approx 0$ bo'lsa, blok osongina identik akslantirishni o'rganadi.

8.1. Ma'lumotni kengaytirish va transfer learning

Augmentatsiya (ma'lumotni kengaytirish). O'qitishda tasvirlarga ma'noni saqlaydigan tasodifiy o'zgartirishlar qo'llanadi: kesish, gorizontaal akslantirish, rang o'zgarishi. Shu tarzda model kerakli invariantliklarni “o'rganadi” va qayta moslashuv (overfitting) kamayadi. Testda augmentatsiya qo'llanmaydi.

Transfer learning: ImageNet da o'qitilgan tarmoqning konvolyutsion qismi universal belgilar chiqaruvchisi sifatida olinadi va faqat yangi bosh o'qitiladi; keyin butun tarmoqni kichik η bilan nozik sozlash (fine-tuning) mumkin.

```

import torch, torch.nn as nn, torchvision
from torchvision import transforms as T
aug = T.Compose([T.RandomResizedCrop(224), T.RandomHorizontalFlip(), # o'qitish
                T.ToTensor(), T.Normalize([0.485, 0.456, 0.406], [0.229, 0.224, 0.225])])
net = torchvision.models.resnet18(weights=None) # amalda: weights="IMAGENET1K_V1"
for p in net.parameters():
    p.requires_grad = False # belgilar muzlatiladi
  
```

```
net.fc = nn.Linear(net.fc.in_features, 5) # yangi bosh: 5 sinf
trainable = sum(p.numel() for p in net.parameters() if p.requires_grad)
print(trainable, net(torch.randn(2, 3, 224, 224)).shape)
opt = torch.optim.AdamW(net.fc.parameters(), lr=1e-3)
# 2565 torch.Size([2, 5]) (512*5 + 5 = 2565 ta o'qitiladigan parametrlar)
```

Amaliy talqin: kichik ma'lumot bilan tasvir klassifikatsiyasi

Korxonada 5 xil mahsulot nuqsoni bo'yicha atigi 2000 ta rasm bor. ResNet-18 ni noldan o'qitish (11,7 mln parametr) kuchli overfitting beradi; muzlatilgan belgilar + yangi bosh esa atigi 2565 parametrlarni o'rganadi va odatda bir necha daqiqada yaxshi natija beradi. Hisoblash byudjeti: ResNet-18 bitta 224×224 tasvir uchun $\approx 1,8$ GMAC — CPU da ham real vaqtga yaqin. Ma'lumot ko'paygan sari yuqori qatlamlarni ham fine-tuning qilish foydali.

Amaliy tanlov

Ma'lumot kam: oldindan o'qitilgan ResNet-18/50 (transfer learning, fine-tuning). Mobil qurilma: MobileNet/EfficientNet (depthwise separable). Yuqori aniqlik va ko'p ma'lumot: ConvNeXt yoki Vision Transformer (5-ma'ruza). Boshlash uchun ResNet-18 — yetarlicha kuchli va sozlash oson.

Tipik xatolar

- PyTorch tensor tartibi $N \times C \times H \times W$; NumPy/PIL tasvirlari esa $H \times W \times C$ — `permute(2,0,1)` unutilsa, o'lchamlar xatosi yoki ma'nosiz natija.
- Baholashda `model.eval()` unutilsa, BN joriy paket statistikasini, dropout esa tasodifiy niqobni ishlatadi — natija paketga bog'liq va yomonroq bo'ladi.
- Oldindan o'qitilgan modelda o'qitishdagi normalizatsiya (ImageNet o'rtachasi va std) bilan bir xil normalizatsiya qo'llanishi shart.
- Flatten dan keyingi Linear kirish o'lchami qo'lda noto'g'ri hisoblanadi — Teorema 3.1 yoki GAP dan foydalaning.
- Augmentatsiya validatsiya/test to'plamiga qo'llanmaydi.

9. Python'da amalga oshirish

1. Chiqish o'lchami va "sodda" konvolyutsiya (NumPy), PyTorch bilan solishtirish.

```
import numpy as np, torch, torch.nn.functional as F
def out_size(h, k, p=0, s=1, d=1):
    return (h + 2*p - d*(k - 1) - 1) // s + 1

def conv2d_naive(X, W, b, s=1, p=0): # X: (C_in, H, W), W: (C_out, C_in, K, K)
    X = np.pad(X, ((0, 0), (p, p), (p, p)))
    C_out, _, K, _ = W.shape
    Ho, Wo = out_size(X.shape[1], K, 0, s), out_size(X.shape[2], K, 0, s)
    Y = np.zeros((C_out, Ho, Wo))
    for o in range(C_out):
        for i in range(Ho):
            for j in range(Wo):
                win = X[:, i*s:i*s+K, j*s:j*s+K] # (C_in, K, K) oyna
                Y[o, i, j] = np.sum(win * W[o]) + b[o]
    return Y

rng = np.random.default_rng(0)
X = rng.normal(size=(3, 9, 9)); W = rng.normal(size=(4, 3, 3, 3)); b = rng.normal(size=4)
for s, p in [(1, 0), (1, 1), (2, 1)]:
    Y = conv2d_naive(X, W, b, s, p)
    Yt = F.conv2d(torch.tensor(X)[None], torch.tensor(W), torch.tensor(b),
                  stride=s, padding=p)[0].numpy()
    print(s, p, Y.shape, f"{np.abs(Y - Yt).max():.1e}")
# 1 0 (4, 7, 7) 3.6e-15 | 1 1 (4, 9, 9) 3.6e-15 | 2 1 (4, 5, 5) 3.6e-15
```

2. Ekvivariantlik tajribasi: CNN va MLP. 16×16 shovqinli tasvirlarda “+” yoki “×” belgisi;

o'qitishda belgi faqat *chap* qismda (ustunlar 0–5), testda faqat *o'ng* qismda (ustunlar 8–13).

```
import numpy as np, torch, torch.nn as nn
P = {0: np.array([[0,1,0],[1,1,1],[0,1,0]]), 1: np.array([[1,0,1],[0,1,0],[1,0,1]])}
def data(n, cols, rng): # 16x16 tasvir: "+" yoki "x" belgisi
    X = 0.3 * rng.normal(size=(n, 1, 16, 16)); y = rng.integers(0, 2, n)
    for k in range(n):
        r, c = rng.integers(0, 14), rng.choice(cols)
        X[k, 0, r:r+3, c:c+3] += P[y[k]]
    return torch.tensor(X, dtype=torch.float32), torch.tensor(y)
rng = np.random.default_rng(0)
Xtr, ytr = data(2000, np.arange(0, 6), rng) # o'qitish: belgi chap qismda
Xte, yte = data(1000, np.arange(8, 14), rng) # test: belgi o'ng qismda
def fit(model, epochs=15):
    opt = torch.optim.Adam(model.parameters(), lr=1e-2)
    for _ in range(epochs):
        for i in range(0, 2000, 50):
            opt.zero_grad()
            nn.functional.cross_entropy(model(Xtr[i:i+50]), ytr[i:i+50]).backward()
            opt.step()
        with torch.no_grad():
            acc = lambda X, y: (model(X).argmax(1) == y).float().mean().item()
            n = sum(p.numel() for p in model.parameters())
            return f'{n}_parametr, {train_{acc(Xtr, ytr):.3f}}, {test_{acc(Xte, yte):.3f}}'
torch.manual_seed(0)
cnn = nn.Sequential(nn.Conv2d(1, 8, 3), nn.ReLU(), nn.AdaptiveMaxPool2d(1),
                    nn.Flatten(), nn.Linear(8, 2))
mlp = nn.Sequential(nn.Flatten(), nn.Linear(256, 32), nn.ReLU(), nn.Linear(32, 2))
print('CNN', fit(cnn)); print('MLP', fit(mlp))
# CNN 98 parametr, train 0.990, test 0.945
# MLP 8290 parametr, train 1.000, test 0.514
```

CNN (98 parametr) test pozitsiyalarini hech qachon ko'rmagan bo'lsa ham 94,5% aniqlik beradi: bitta filtr hamma joyda ishlaydi (Teorema 2.4) va global max-pooling joylashuvni “unutadi”. MLP (8290 parametr) o'quv to'plamini 100% yodlaydi, lekin testda 51,4% — tasodifiy taxmin darajasi: o'ng qismdagi piksellar uchun og'irliklar deyarli o'rganilmagan. Parametrlar soni emas, *to'g'ri induktiv siljish* hal qiluvchi.

3. Residual blok (PyTorch).

```
import torch, torch.nn as nn
class ResidualBlock(nn.Module):
    def __init__(self, c):
        super().__init__()
        self.body = nn.Sequential(
            nn.Conv2d(c, c, 3, padding=1, bias=False), nn.BatchNorm2d(c), nn.ReLU(),
            nn.Conv2d(c, c, 3, padding=1, bias=False), nn.BatchNorm2d(c))
    def forward(self, x):
        return torch.relu(x + self.body(x)) # identiklik yo'li: x + F(x)
blk = ResidualBlock(64)
print(sum(p.numel() for p in blk.parameters()), blk(torch.randn(2, 64, 56, 56)).shape)
# 73984 torch.Size([2, 64, 56, 56])
```

Parametrlar: $2 \cdot (64 \cdot 64 \cdot 9) + 2 \cdot (2 \cdot 64) = 73\,728 + 256 = 73\,984$ (konvolyutsiyalarda siljish yo'q, BN da γ, β). Kirish va chiqish o'lchami bir xil — aks holda qisqa yo'lga 1×1 konvolyutsiya qo'yiladi.

10. Xulosa

1. CNN uchta farazga tayanadi: lokallik, og'irliklarni bo'lishish, ierarxiya; natijada parametrlar keskin kamayadi.
2. Kutubxonalardagi konvolyutsiya — kross-korrelatsiya; $s = 1$ da u siljishga ekvivalent.
3. $H_{\text{out}} = \lfloor (H + 2p - d(K - 1) - 1)/s \rfloor + 1$; parametrlar $C_{\text{out}}(C_{\text{in}}K^2 + 1)$.
4. 1×1 va depthwise separable konvolyutsiyalar hisobni bir necha marta kamaytiradi.
5. Pooling o'lchamni kamaytiradi, lekin to'liq siljish invariantligini bermaydi; qabul maydoni rekursiv

hisoblanadi.

6. Konvolyutsiya gradientlari ham konvolyutsiyalar: $\nabla W = X \star \delta$, ∇X — ag'darilgan yadro bilan to'liq konvolyutsiya.
7. Batch normalization va residual bog'lanishlar chuqur CNN larni o'qitish imkonini beradi.

11. Nazorat savollari

1. Nega to'liq bog'langan qatlam tasvir uchun samarasiz?
2. CNN ning uchta asosiy farazi nima?
3. Konvolyutsiya va kross-korelyatsiya farqi nimada?
4. Ko'p kanalli konvolyutsiya formulasini yozing.
5. Ekvivariantlik va invariantlik farqi nima?
6. Chiqish o'lchami formulasini keltirib chiqaring.
7. "Same" padding qachon o'lchamni saqlaydi?
8. Konvolyutsion qatlam parametrlari qanday sanaladi?
9. 1×1 konvolyutsiya nima qiladi?
10. Depthwise separable konvolyutsiya qanday tejaydi?
11. Max-pooling orqali gradient qanday o'tadi?
12. Qabul maydoni qanday hisoblanadi?
13. Nega ikkita 3×3 bitta 5×5 dan afzal?
14. Konvolyutsiya filtri gradienti qanday topiladi?
15. im2col g'oyasi nima?
16. Batch normalization o'qitish va inferensiyada qanday farq qiladi?
17. Residual bog'lanish gradientga nima beradi?
18. VGG-16 parametrlarining asosiy qismi qayerda?

12. Mashqlar (javoblari bilan)

A guruh.

1. $H = 32$, $K = 5$: (a) $p = 0$, $s = 1$; (b) $p = 2$, $s = 2$. *Javob:* 28; 16.
2. $3 \rightarrow 16$ kanal, 5×5 filtrli qatlam parametrlari. *Javob:* $16(75 + 1) = 1216$.
3. $\begin{pmatrix} 1 & 3 & 2 & 0 \\ 4 & 2 & 1 & 1 \\ 0 & 1 & 5 & 2 \\ 2 & 2 & 3 & 4 \end{pmatrix}$ ga 2×2 , $s = 2$ max va average pooling. *Javob:* $\begin{pmatrix} 4 & 2 \\ 2 & 5 \end{pmatrix}$; $\begin{pmatrix} 2,5 & 1 \\ 1,25 & 3,5 \end{pmatrix}$.
4. $H = 32$, $K = 3$, $p = 1$, $s = 2$ qatlam ikki marta ketma-ket. *Javob:* $32 \rightarrow 16 \rightarrow 8$.
5. $H = 28$, $K = 3$, $p = 0$, $s = 1$, $d = 3$. *Javob:* $K_e = 7$, $H_{out} = 22$.
6. $224 \times 224 \times 3$ tasvirni 1000 neyronga to'liq ulash parametrlari (siljish bilan). *Javob:* 150 529 000.

B guruh.

- , conv3, conv3 qabul maydoni. *Javob:* 14.
7. Barcha $s = 1$ bo'lgan beshta ketma-ket 3×3 qatlam qabul maydoni. *Javob:* 11.
 8. Depthwise separable qatlam (3×3 , $64 \rightarrow 128$): parametrlar va oddiy qatlamga nisbat. *Javob:* 8960 ; $73856/8960 \approx 8,2$.
 9. BN: $(1, 2, 3, 6)$, $\gamma = 2$, $\beta = 1$, $\epsilon \rightarrow 0$. *Javob:* $\approx (-1,138; -0,069; 1; 4,207)$.
 10. 2.2-misolda $L = \sum_{i,j} G_{ij} Y_{ij}$, $G = K$ bo'lsa, $\partial L / \partial K$ va $\partial L / \partial X$. *Javob:* $\begin{pmatrix} 0 & -1 \\ -1 & 1 \end{pmatrix}$; $\text{diag}(1, -2, 1)$ (qolganlari 0).
 11. Chiqishi $28 \times 28 \times 32$, kirishi 16 kanal, 5×5 qatlam uchun MAC. *Javob:* $28^2 \cdot 32 \cdot 16 \cdot 25 = 10\,035\,200$.

C guruh.

13. 1D misolda $s = 2$ konvolyutsiya bir pozitsiyali siljishga ekvivalent emasligini ko'rsating. *Ko'rsatma:* $K = (1)$, $s = 2$: $[1, 0, 0, 0] \rightarrow [1, 0]$, $[0, 1, 0, 0] \rightarrow [0, 0]$.
14. Teorema 6.1 dagi $\partial L / \partial X$ formulasini $s = 2$ uchun umumlashtiring.
15. VGG-16 da FC qatlamlarni GAP + Linear(512, 1000) bilan almashtirsa, parametrlar qancha bo'ladi? *Javob:* $14\,714\,688 + 513\,000 = 15\,227\,688$ (≈ 9 marta kam).
16. Uchta 3×3 va bitta 7×7 qatlam ($C \rightarrow C$) og'irliklarini solishtiring. *Javob:* $27C^2$ va $49C^2$.
17. $J_l = 0,5$ ($l = 1, \dots, 10$) bo'lsa, oddiy va residual tarmoqda gradient ko'paytuvchisi. *Javob:* $0,5^{10} \approx 9,8 \cdot 10^{-4}$; $1,5^{10} \approx 57,7$ (portlash xavfi — shuning uchun BN, kichik boshlang'ich F).
18. conv2d_naive ga dilation va groups parametrlarini qo'shing va PyTorch bilan solishtiring.

Adabiyotlar

- [1] Goodfellow I., Bengio Y., Courville A. *Deep Learning*. MIT Press, 2016 (9-bob).
- [2] LeCun Y., Bottou L., Bengio Y., Haffner P. Gradient-based learning applied to document recognition. *Proc. IEEE*, 86(11), 1998.
- [3] Krizhevsky A., Sutskever I., Hinton G. ImageNet classification with deep convolutional neural networks. *NeurIPS*, 2012.

- [4] Simonyan K., Zisserman A. Very deep convolutional networks for large-scale image recognition. *ICLR*, 2015.
- [5] He K., Zhang X., Ren S., Sun J. Deep residual learning for image recognition. *CVPR*, 2016.
- [6] Ioffe S., Szegedy C. Batch normalization: accelerating deep network training by reducing internal covariate shift. *ICML*, 2015.
- [7] Dumoulin V., Visin F. A guide to convolution arithmetic for deep learning. arXiv:1603.07285, 2016.
- [8] Howard A. G. va boshq. MobileNets: efficient convolutional neural networks for mobile vision applications. arXiv:1704.04861, 2017.